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The reorientation mechanism of hydroxide ions in water: A molecular dynamics study

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ABSTRACT

Classical molecular dynamics simulations with a polarizable force field indicate two major structural motifs for the aqueous bulk solvation of OH^- : a four- and a five-coordinated solvent hydrogen bond donor to OH^- in addition to a weak solvent hydrogen bond acceptor by OH^- . A two-step mechanism for the reorientation of OH^- in water is proposed: first, the reorientation of OH^- is initiated by the coupled translation with the water molecules in its first solvation shell; second, the OH^- relaxes to the minimum energy configuration. The first step is the rate-limiting one for this mechanism.

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1. Introduction

As one of the two ions formed by water dissociation, hydroxide plays a unique role in chemical and biochemical processes. In contrast to the well-established mechanism for the mobility of hydronium ions in water [1–4], the debate over the mechanism of the orientation relaxation of hydroxide ions in solution is still quite active [5–12]. The traditional view of the transport mechanism for both hydronium and hydroxide ions is similar [5,7], namely that the transport of both ions is driven by second solvation shell fluctuations. By breaking one hydrogen bond in the second solvation shell, the coordination number of a first solvation shell water is reduced from 4 to 3 and this under-coordinated hydrogen bonded network will subsequently form either a $[\text{H}_2\text{O}-\text{H}-\text{OH}_2]^+$ complex with hydronium or a $[\text{HO}-\text{H}-\text{OH}]^-$ complex with hydroxide to facilitate the proton transfer between those ions in water. This mechanism is based on a tri-coordinated structure of hydroxide ions that is similar to the structure of hydronium ions. This triply hydrogen bonded complex $\text{OH}^-(\text{H}_2\text{O})_3$ with a tetrahedral arrangement is supported by some theoretical and experimental studies [11,13].

Previous quantum calculations by Tuckerman et al. suggested a roughly planar arrangement of a fourfold coordinated $\text{OH}^-(\text{H}_2\text{O})_4$ complex being the most probable arrangement in liquid water, whereas $\text{OH}^-(\text{H}_2\text{O})_3$ is a transition state during the proton transfer process [6,8,9,12]. This suggestion leads to the following picture of the transport mechanism of OH^- in water: one hydrogen bond in the ion's first solvation shell is broken due to thermal fluctuations; the remaining threefold coordinated complex $\text{OH}^-(\text{H}_2\text{O})_3$ donates its hydrogen to form a weak hydrogen bond with a neighboring water; the proton transfer between OH^- and one of the first solva-

tion shell water molecules is then completed, causing OH^- and H_2O to exchange places. The fourfold coordinated hydroxide ion moiety is further supported by several recent experiments [14–16] together with a weak hydrogen bond in which the H atom of OH^- acts as a donor to a fifth water molecule [14,17,18].

Although progress into the structural and dynamical properties of OH^- in aqueous solutions has been previously reported, the direct measurement of the rotation of OH^- was only recently made possible due to complications with the experimental techniques [19]. Thøgersen et al. used the charge-transfer-to-solvent (CTTS) transition of hydroxide ions in water to study the reorientation of hydroxide ions at various temperatures. From their observations, the reorientation of hydroxide ions is well described by a single exponential decay function after 3 ps. At temperatures above 290 K, the hydroxide ions have a similar orientational relaxation time as that of water molecules, while at temperatures below 290 K, the relaxation time of hydroxide ions is much longer than that of the water molecules. The authors proposed two different mechanisms for this process for the different temperature ranges. At high temperatures, the reorientation is initiated by the movement of a second solvation shell water into the first solvation shell of the hydroxide ion, and the over-coordinated hydroxide breaks one of its hydrogen bonds and undertakes a large angular jump. At low temperatures, the stronger interaction between hydroxide and its first solvation shell water molecules creates a type of super-complex that has an increased moment of inertia, which slows down the reorientation of the hydroxide ions dramatically. Furthermore, they suggested the possibility of a change in the hydration structure motif from $\text{OH}^-(\text{H}_2\text{O})_3$ at high temperatures to $\text{OH}^-(\text{H}_2\text{O})_4$ at low temperatures.

In this Letter, we report a detailed investigation of the reorientation mechanism of hydroxide ions using molecular dynamics (MD) simulations with polarizable potentials that have been previously widely applied to understand the structural and dynamical

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properties of liquids [20]. Polarizable potentials in classical MD simulations have been shown to be essential in describing the behavior of anions, especially at the interfaces [21–24], but also in bulk water [25]. A general deficiency of classical rigid molecule potentials is their inability to account for proton transfer between molecules. However, for the present study the adopted approach is justified for two reasons. First, the decay of the reorientation of the hydroxide ions after 3 ps can be well described by a single exponential function, a fact that lends strong evidence to the proposition that the mechanism is facilitated via a succession of small angular jumps [26]. Second, the proton transfer rate is believed to occur in 3–4 ps at room temperature. At lower temperatures, it is reasonable to assume the rate will decrease. This time scale is longer than the experimental observed relaxation time of the reorientation for hydroxide ions, thus the reorientation of hydroxide ions is little affected by the proton transfer. We also present a critical comparison against high-level electronic structure results for the coordination number of hydroxide in small aqueous clusters with the polarizable force field used in this study and other DFT-based approaches that were previously adopted to study the aqueous solvation of OH^- .

2. Computational details

The polarizable Dang–Chang (DC) force field for water, potassium and hydroxide ions were used in the MD simulations. These interaction potentials are described in earlier work [27]. We used a cubic periodic simulation box comprised of 1000 water molecules and 9 KOH ion pairs, corresponding to a concentration of 0.5 M for KOH. Simulations at lower concentrations (0.05 and 0.25 M) of KOH were also carried out. The Shake algorithm was used to keep the O–H bonds rigid [28]. The Lennard–Jones interactions were truncated at 11 Å, and long range electrostatic interactions were evaluated using the Particle mesh 3D periodic Ewald summation [29]. The time step was 1 fs for all simulations. The systems were equilibrated for 3 ns in the isobaric–isothermal (NPT) ensemble and subsequent simulations for data collection were carried out in the micro-canonical (NVE) ensemble.

The electronic structure calculations for the various isomers of the $\text{OH}^-(\text{H}_2\text{O})_n$, $n = 4$ and 5 clusters were performed at the second order Møller–Plesset (MP2) [30] and coupled-cluster singles and doubles with perturbative estimate of triple excitations [CCSD(T)] [31,32] levels of theory using the family of augmented correlation consistent basis sets, aug-cc-pVnZ ($n = \text{D, T and Q}$), of Dunning and co-workers [33,34]. At the MP2 level of theory the geometries of the 4-coordinated (C_4 symmetry) and 3-coordinated ('3 + 1') $\text{OH}^-(\text{H}_2\text{O})_4$ cluster isomers were optimized with up to aug-cc-pVQZ basis set while CCSD(T) single point energy calculations were run at the MP2 optimal geometries. For the $\text{OH}^-(\text{H}_2\text{O})_5$ cluster the geometries of the 5-coordinated (C_5 symmetry), 4-coordinated ('4 + 1') and 3-coordinated ('3 + 1') isomers were optimized with the aug-cc-pVDZ and aug-cc-pVTZ basis sets. For all the previous isomers harmonic frequency calculations were also performed to ensure that they correspond to minimum configurations. The MP2 and CCSD(T) calculations were carried out with the NWChem suite of electronic structure codes [35,36].

Additional cluster optimizations with the Perdew–Burke–Ernzerhof (PBE) [37] and the Becke–Lee–Yang–Parr (BLYP) [38,39] density functionals for the above isomers of the hydroxide ion with 4 and 5 water molecules were performed in order to compare the performance of those functionals against the more structures (MP2) and relative energetics [MP2, CCSD(T)] obtained from the higher level calculations. The BLYP cluster optimizations were performed with both a plane-wave (Martins–Troullier pseudopotentials with a wavefunction energy cutoff of 70 Ry) and with a hybrid Gaussian and plane-wave (GPW) method. For the GPW

method a unit cell of 15 Å was used, the core electrons were removed by the introduction of norm-conserving pseudopotentials developed by Goedecker, Teter and Hutter (GTH) [40,41], the Kohn–Sham orbitals were expanded in a triple-zeta valence basis set augmented with two sets of d -type or p -type polarization functions (TZV2P) while a charge density cutoff of 280 Ry was used for the auxiliary basis set. The same basis set (TZV2P, 280 Ry cutoff) was used for the GPW optimizations with the PBE functional. The GPW optimizations were performed with Quickstep [42], which is part of the Cr2k program package, whereas the BLYP/plane-wave optimizations were performed with the CPMD program package [43].

3. Results and discussion

3.1. Reorientation relaxation times

The relaxation time of the reorientation of hydroxide ions was calculated from the orientation correlation function $C(t)$, defined as

$$C(t) = \langle P_2(\mathbf{u}(t) \cdot \mathbf{u}(0)) \rangle, \quad (1)$$

where $\mathbf{u}$ is the unit vector along the O–H direction, and P_2 is the second-order Legendre polynomial. The relaxation time is given by the integral,

$$\tau = \int_0^\infty C(t) dt. \quad (2)$$

For both OH^- and H_2O at different temperatures, the decay of the orientational correlation function is described by a single exponential. This observation indicates a successive small angular jump mechanism for the reorientation of both OH^- and H_2O [26]. At low temperatures, $C(t)$ shows an anomalous slow decay that was also observed in the CTTS experiments [19]. Fig. 1 shows the computed relaxation time of the hydroxide ion reorientation as a function of temperature. The computed relaxation time increases with decreasing temperature in qualitative agreement with the CTTS experimental data.

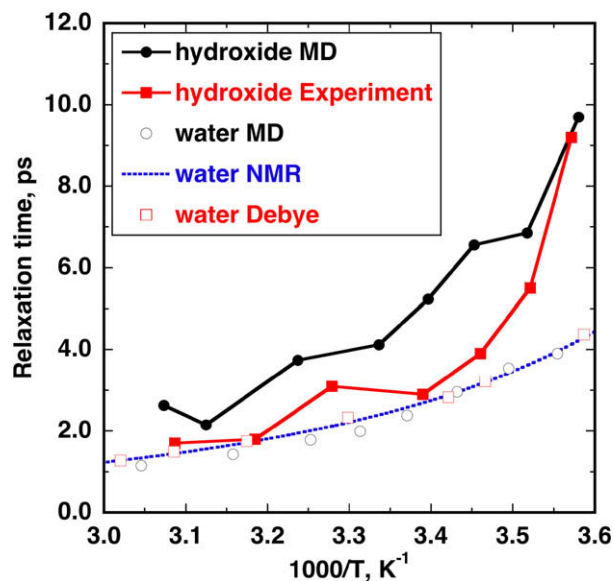


Fig. 1. The computed relaxation time of the hydroxide and water reorientation as a function of temperature. Black filled circles are the hydroxide simulation results; red filled squares are the hydroxide CTTS experimental results. Black open circles are the water simulation data; blue dotted line is the water NMR ($\tau_{\text{NMR}} = 0.34 \times (T_{233}^{-1.83})$) and red open squares are water THz measurements ($\tau_{\text{Debye}}/3$).

The reorientation of water was also previously investigated using the same experimental techniques. The two experimentally available orientational relaxation times for water (from NMR [19,44] and THz [45] measurements) are shown in Fig. 1 for comparison. As described in the experimental paper, the THz results were obtained by dividing the Debye time by a factor of 3 with the assumption that the reorientation of water follows a successive small angular jump mechanism [19]. Our simulation results are in excellent agreement with the experiments over the temperature range shown in Fig. 1, and they provide further support the successive small angular jump mechanism for the reorientation of water, although this is still a subject of future research [46].

Although the relaxation times for the reorientation for both hydroxide ions and water molecules increase with decreasing temperature, a large difference in their corresponding rates of change with temperature was observed. At $T > 290$ K the reorientation relaxation times of the hydroxide ion and water are similar, however, at $T < 290$ K the reorientation of the hydroxide ions slows down more dramatically when compared to that of the water molecules, in agreement with the CTTS experiments [19]. Two different reorientation mechanisms of hydroxide ions for high and low temperatures were proposed based on the assumption that the dominant solvation structures of hydroxide ions changes with temperature [19]. It is therefore necessary to investigate the solvation structure of hydroxide ions at different temperatures.

3.2. Solvation motifs

To gain more insight into the solvation structure of hydroxide ions, we calculated the number of hydrogen bonds (HB) formed by the hydroxide ions with surrounding water molecules. We used the following geometric criteria to define a HB: the distance between the O of OH^- and the O of H_2O $R_{\text{OO}} < 3.5$ Å, the hydrogen bonded distance $R_{\text{O}\cdots\text{H}} < 2.6$ Å and finally the hydrogen bonded angle $\gamma(\text{O}\cdots\text{H}\cdots\text{O}) > 125^\circ$. The total number of HB as a function of temperature is shown in Fig. 2 (black circles). As expected, the total number of HBs is not temperature sensitive, with the average number of HBs being ~ 5 . This number is slightly larger than the coordination number in the first solvation shell (4.73), implying that a weak transient HB is formed between the OH^- and the H_2O in the second solvation shell of the OH^- . In Fig. 2, the total number

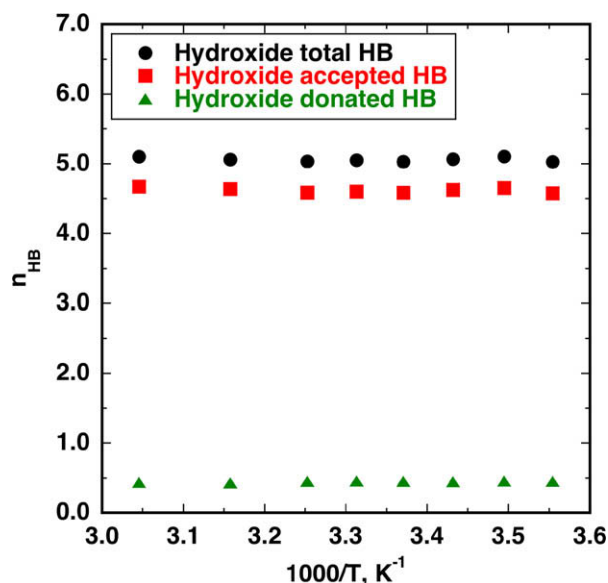


Fig. 2. The total (black circles), acceptor (red squares) and donor (green triangles) HBs for the hydroxide ion as a function of temperature.

of HBs is decomposed into the number of HBs accepted by OH^- (red squares), and donated by OH^- (green triangles). As with the total number of HBs, no temperature dependence was observed for those as well. The average number of HBs accepted and donated by OH^- (subject to the geometric criteria for defining a HB listed earlier) is ~ 4.6 and ~ 0.4 , respectively. These numbers indicate the existence of more than one single dominant solvation motif for OH^- . For the HBs accepted by OH^- , the average probability of finding 4 HBs is $\sim 38\%$, and that for 5 HBs is $\sim 57\%$.

The OH^- is also capable of donating its hydrogen to participate in a weak hydrogen bond with the oxygen of a water molecule. Our simulations suggest two major solvation structures of OH^- having different numbers of OH^- acceptor HBs. These solvation structures are shown in Fig. 3. In structure (a) the hydroxide ion has four acceptor HBs from the water molecules; these four water molecules adopt a nearly planar arrangement with the center of the plane below the O of the hydroxide ion. The orientation of the hydroxide ion is found more likely to be perpendicular to the plane of the 4 O atoms of the water molecules in order to avoid the repulsion between the H atoms of OH^- and H_2O . A weak HB can be formed by a water molecule approaching OH^- as an acceptor from the top. The quasi-planar arrangement of the four water molecules in the first solvation shell of the hydroxide ion is supported by previous quantum mechanical simulations [6,9,14], as well as neutron diffraction [14], X-ray adsorption [15] and diffraction experiments [16]. In addition, the hydroxide donating HB was also previously reported in both theoretical and experimental studies [9,14,17,18]. In structure (b) of Fig. 3, the hydroxide accepts five HBs from the surrounding water molecules. Four of the waters are arranged in a manner similar to the quasi-planar structure (a). Compared to structure (a), the distance between the hydroxide oxygen and the center of the water plane is closer in structure (b). The O atom of hydroxide is closer to the plane defined by the O atoms of the four water molecules. This kind of arrangement results in a nearly tetragonal pyramid that has a smaller height in structure (b) than in structure (a) and allows more space for an additional water molecule to approach underneath in order to form the fifth donating HB to OH^- . In concert, the repulsion from the four planar water molecules to the other water molecules on the donating side of OH^- is increased, and thus the probability of forming a donor HB for the hydroxide is reduced. Structure (b) was found in a recent CPMD investigation of alkaline solutions [8]. Similar to structure (a), the orientation of the hydroxide ion is more likely to be perpendicular to the plane defined by the center of mass of the four water molecules.

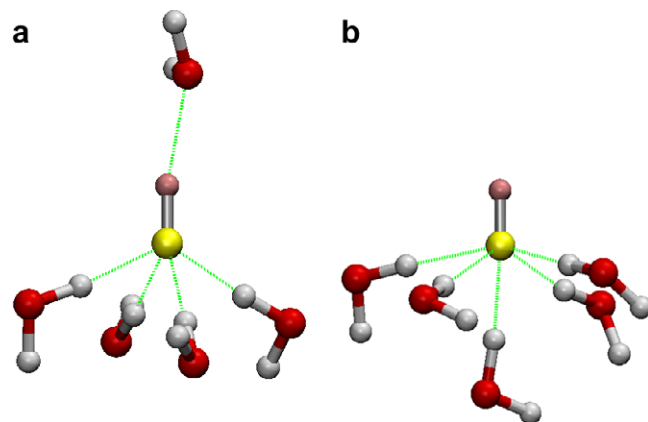


Fig. 3. The two dominant $\text{OH}^-/\text{H}_2\text{O}$ complex structures found in our simulation. The yellow and pink spheres correspond to the O and H atoms of OH^- , whereas the red and white spheres to the O and H atoms of H_2O , respectively. The HBs are denoted by green dotted lines.

The four-coordinated hydroxide ion has a slightly different structure when compared to previous DFT simulations of Tuckerman et al. [6,9,12]. In this study, the observed distance between the center of plane formed by four water molecules and the oxygen in hydroxide ion is larger than that observed in the previous DFT simulations, and the four water molecules around the hydroxide ion are in a more distorted planar arrangement than in the previous DFT simulations. However, the arrangement of the four planar water molecules in the five-coordinated hydroxide ion complex is similar to the results of previous DFT simulations. Since the O–O coordination numbers are similar in both simulations, ~ 4.7 and ~ 4.5 for classical and quantum, respectively, the disagreement of the number of HB formed by hydroxide ion in the current and DFT studies could be due to the definition of the hydrogen bond in this work.

3.3. HB dynamics

The number of HBs accepted by hydroxide as a function of time at different temperatures is shown in Fig. 4 at $T = 281$ K (top panel) and $T = 302$ K (bottom panel). Clearly, four and five hydroxide acceptor HBs correspond to the dominant structures. There are some transient formations of three and six HBs due to thermal fluctuations. Although the population of four and five hydroxide acceptor arrangements is roughly the same at the two different temperatures, the switching rate between these two structures is temperature dependent. In general, this switching rate is lower as the temperature decreases. At $T = 281$ K, the lifetime of one structure, especially when there are five acceptor HBs, can be as long as 10 ps. In contrast, at $T = 302$ K the breaking and making of hydrogen bonds is much faster, the typical time of effective switching is now under 3 ps. For both temperatures, oscillations between the two structures are observed.

Fig. 5 is a schematic illustration of one full cycle of structure switching. The yellow and pink spheres correspond to the O and H atoms of OH^- , whereas the red and white spheres to the O and

H atoms of H_2O , respectively, and the HBs are represented by green dotted lines. In Fig. 5a, the system is in the configuration of five hydroxide acceptor HBs. Due to the thermal fluctuations, one of those five HBs is broken and the system changes to the four acceptor HB configuration shown in Fig. 5b. Usually at the beginning of the switching process, the donating to OH^- water molecules are not close to the quasi-planar arrangement, and the relaxation to a more energetically stable arrangement takes place after the formation of the new structure. In Fig. 5c, the quasi-planar arrangement of the four donating water molecules is again distorted by thermal fluctuations to generate more space for a new HB to be formed by the hydroxide ion with another water molecule that was initially in the second solvation shell. Then, a former first solvation shell water molecule moves to the second solvation shell of the hydroxide ion. Finally, the newly formed structure relaxes to its stable configuration shown in Fig. 5d. Although the switching between the structures involves the breaking and forming of HBs, no drastic changes in the overall arrangement are observed. The driving force for these structural transitions is the thermal fluctuations. The system has to be in a specific geometry for the breaking or forming of HBs to facilitate the processes.

Since the interaction between the hydroxide ion and its first solvation shell water molecules is strong, the motion of the hydroxide ion is coupled to the motion of its first solvation shell water molecules. As stated earlier, the orientation of the hydroxide ion is likely to be perpendicular to the four-water plane. When that plane is distorted due to thermal fluctuations, the hydroxide ion will reorient accordingly. According to our simulations, a two-step mechanism for the reorientation of the hydroxide ion in water is suggested. First, the thermal fluctuation of H_2O in the first solvation shell of OH^- drags the oxygen of OH^- to translate together with the H_2O molecules that are donating a HB to OH^- , initiating the reorientation of OH^- . Second, the hydroxide ion reorients its H to the minimum energy arrangement in the $\text{OH}^-(\text{H}_2\text{O})_n$ complex to complete the reorientation of the OH^- . The second step is very fast due to the small mass of the hydrogen. However, sometimes the reorientation of the OH^- hydrogen involves breaking or forming a weak HB donated by OH^- to other water molecules. This step could be slowed down or accelerated, but the thermal averaging is canceling that out. The rate-limiting step in this mechanism is the first step, namely the thermal fluctuations of the first solvation shell water molecules. Here, the thermal fluctuations of water molecules are related to the translational diffusion of the water molecules.

The translational diffusion of water molecules is investigated from the diffusion coefficient

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} \left\langle \frac{1}{N} \sum_i |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle. \quad (3)$$

Here $\mathbf{r}_i(t)$ is the position of particle i at time t , and N is the total number of particles. At $T = 302$ K, the diffusion coefficient of water obtained from our simulations was $1.9 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$. Compared to the experimental value of $2.3 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ at 298 K [47], our potential was found to slightly underestimate the translation rate of water molecules. This could be one of the explanations for the overestimation of the reorientation relaxation time of hydroxide ions, since the reorientation of hydroxide ions is coupled to the translation diffusion of water molecules in our suggested mechanism. It should be noted that our force field should be considered as an effective one in which quantum effects (known to increase the diffusion coefficient by as much as 60%) [48,49] are implicitly contained, since the intent was to reproduce macroscopic properties of liquid water with classical simulations. The diffusion of water molecules decreases drastically with decreasing temperature. This is consistent with the observation of the drastic slowdown of the reorientation of hydroxide ions at low temperatures. The activation

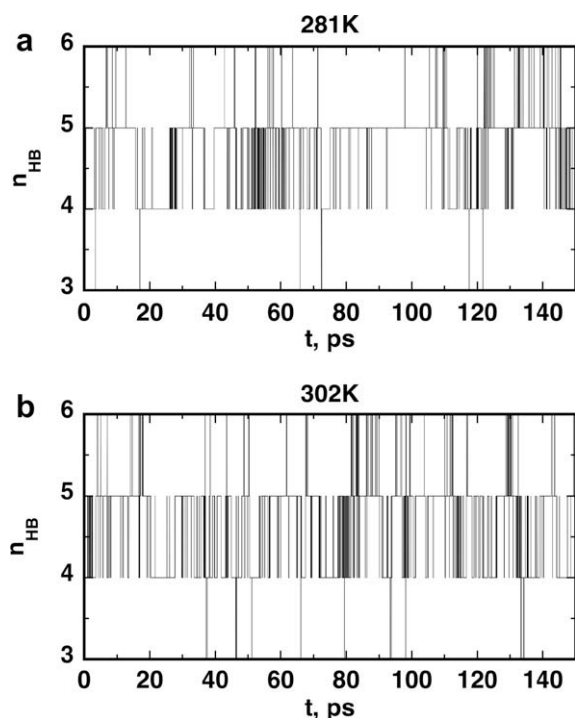


Fig. 4. The number of hydroxide acceptor HBs during the simulation at $T = 281$ K (top panel) and $T = 302$ K (bottom panel).

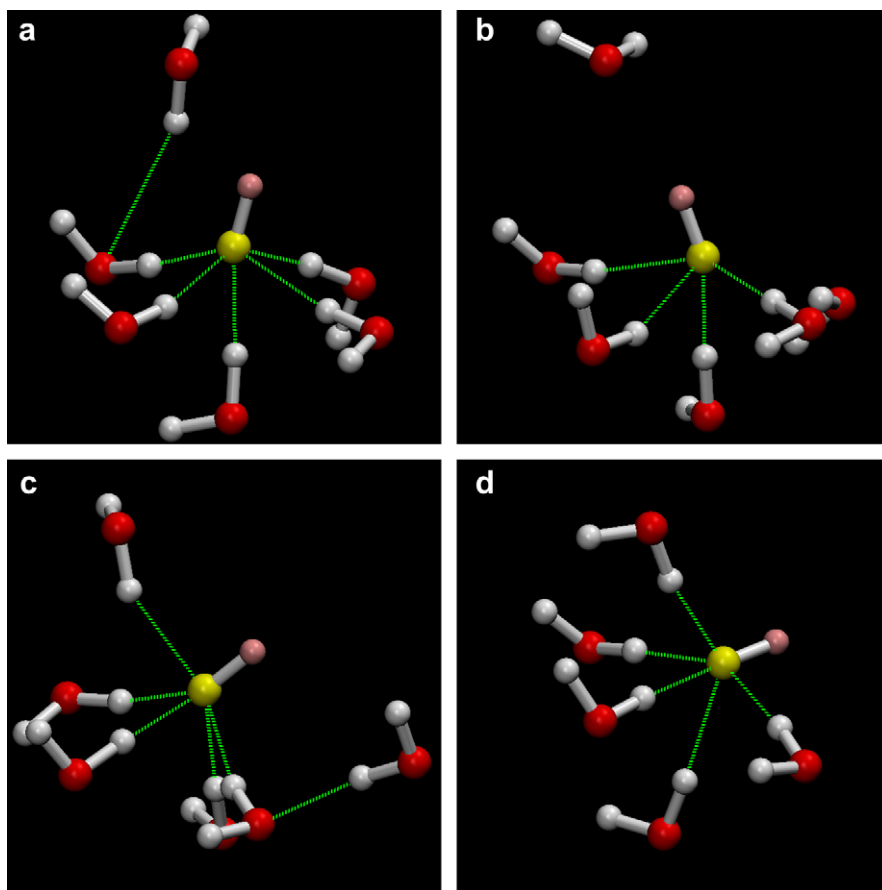


Fig. 5. A schematic illustration demonstrating a full cycle of the structure switching. The atoms have the same color scheme as in Fig. 3.

energies of reorientation of OH^- and translation of H_2O determined from our simulations are 22.5 and 18.5 kJ/mol, respectively. The activation energy for the diffusion in the temperature range of this study can be described by Arrhenius law, $D = D_0 \exp(-E_{at}/RT)$. Here E_{at} is the translational (diffusion) activation energy, R is the gas constant and D_0 is the pre-exponential factor. Similarly, the activation energy for rotation can be described by $\frac{1}{\tau} = \frac{1}{\tau_0} \exp(-E_{ar}/RT)$, where E_{ar} is the rotational activation energy. This lends more evidence that the translational diffusion of water molecules is the dominant step in our proposed two-step mechanism for the reorientation of OH^- . The 4 kJ/mol difference in the activation energies between the translation of H_2O and the reorientation of OH^- is mainly due to the second step in the mechanism, namely the reorientation of the H atom of OH^- .

3.4. The coordination of hydroxide in small water clusters

The previous discussion has made it evident that the accurate description of the different coordination of the first few water molecules acting as donors to hydroxide and their variation with temperature is essential in order to probe its dynamical properties in an aqueous environment. A zero-order picture can be obtained from the relative energetics of the various isomers of the first few hydroxide–water clusters $\text{OH}^-(\text{H}_2\text{O})_n$ corresponding to different coordination of the first solvation shell [50]. In our opinion, the accurate description of the relative coordination of hydroxide in gas phase clusters for a given theoretical approach represents a ‘necessary but not sufficient’ condition for obtaining a rational picture of its local environment in the condensed phase. Furthermore, the comparison between DFT-based approaches and high-level electronic structure results is justified, as they both represent

different levels of approximations to the Born–Oppenheimer potential energy surface (PES). The optimal geometries of the $\text{OH}^-(\text{H}_2\text{O})_n$, $n = 4$ and 5 cluster isomers corresponding to the different coordination of the water molecules are shown in Fig. 6. They all represent minima on the corresponding PESs, as evidenced from the analysis of the vibrational frequencies.

For the $n = 4$ cluster, we examined the four-coordinated isomer of C_4 symmetry and the three-coordinated version with the fourth water molecule in the second solvation shell (denoted as ‘3 + 1’). For the $n = 5$ cluster, we examined the five-coordinated (C_5), four-coordinated (‘4 + 1’) and three-coordinated (‘3 + 2’) isomers, respectively. The results for the relative energetics between those isomers are listed in Table 1. For $n = 4$, the four-coordinated isomer is lower in energy than the three-coordinated (3 + 1) one, although their energy difference is small and it is decreasing upon the use of a larger basis set at the MP2 level of theory. The effect of higher correlation, estimated at the CCSD(T) level of theory with the aug-cc-pVTZ basis set, was found to further stabilize the C_4 isomer by 0.26 kcal/mol with respect to MP2. Based on the results listed in Table 1, we estimate an energy separation (ΔE_e) of just 0.09 kcal/mol at the MP2 Complete Basis Set (CBS) limit and 0.35 kcal/mol at the CCSD(T)/CBS limit, with the C_4 isomer being more stable. The inclusion of harmonic zero-point energy corrections (estimated at the MP2/aug-cc-pVDZ level) to the CCSD(T)/CBS value decreases the energy separation to 0.02 kcal/mol. However, estimates for the enthalpy and Gibbs free energy (albeit using harmonic corrections) stabilize the 3 + 1 isomer with respect to the C_4 by 0.18 and 0.48 kcal/mol, respectively. These results suggest that the energy difference between the two cluster isomers (four- and three-coordinated hydroxide) is quite subtle. The near degeneracy, predicted by the high-level [MP2 and CCSD(T)] calculations, is not

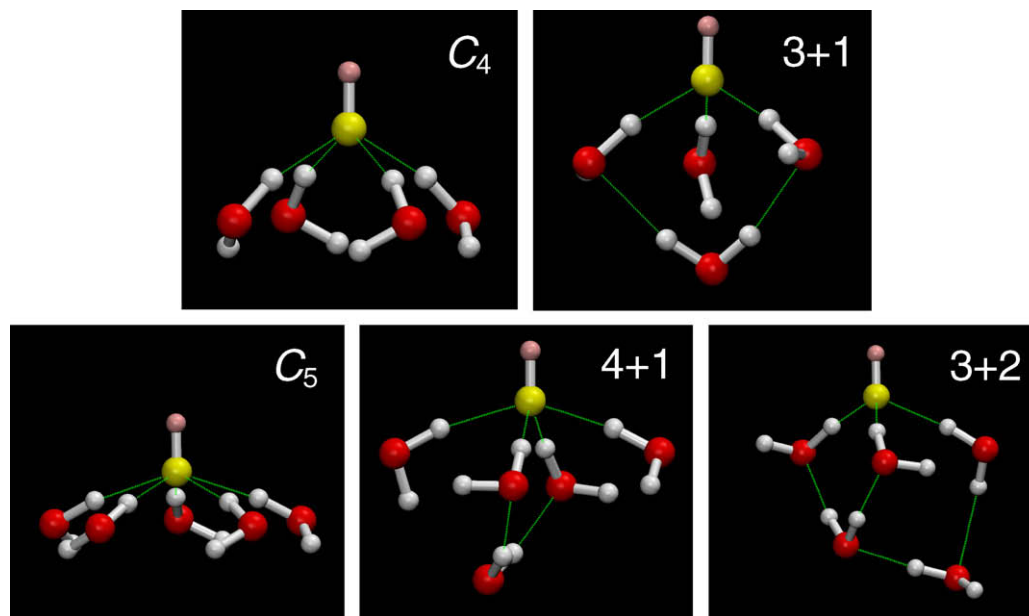


Fig. 6. The various isomers of the $\text{OH}^-(\text{H}_2\text{O})_n$, $n = 4$ and 5 clusters corresponding to the different coordination of hydroxide in the first solvation shell.

Table 1

The relative energetics (in kcal/mol) between the various $\text{OH}^-(\text{H}_2\text{O})_n$, $n = 4$ and 5 isomers.

Method	Quantity	$\text{OH}^-(\text{H}_2\text{O})_4$		$\text{OH}^-(\text{H}_2\text{O})_5$		
		C_4	3 + 1	C_5	4 + 1	3 + 2
MP2/aug-cc-pVDZ	ΔE_e	0	0.56	3.40	0	3.44
MP2/aug-cc-pVTZ	ΔE_e	0	0.15	3.24	0	3.24
MP2/aug-cc-pVQZ	ΔE_e	0	0.10	3.03 ^a	0	3.20 ^a
CCSD(T)/aug-cc-pVTZ ^a	ΔE_e	0	0.41	–	–	–
BLYP/Martens–Troullier/70 Ry	ΔE_e	0	–1.85	3.91	0	0.43
BLYP/TZV2P/280 Ry	ΔE_e	0	–1.15	3.79	0	1.05
PBE/TZV2P/280 Ry	ΔE_e	0	–1.22	4.38	0	1.07
MP2/CBS	ΔE_e (0 K)	0	0.09			
CCSD(T)/CBS + ZPE (MP2/aug-cc-pVDZ)	ΔE_e (0 K)	0	0.35			
	ΔE_0 (0 K)	0	0.02			
	ΔH (298 K)	0	–0.18			
	ΔG (298 K)	0	–0.48			
DC force field	ΔE (0 K)	0	1.30	4.82	0	1.67

^a At the MP2/aug-cc-pVTZ optimal geometry.

well described by the two DFT functionals, which tend to favor the threefold coordinated arrangement over the fourfold one, whereas additional corrections for the zero-point and thermal effects will probably increase their energy separation even more. In contrast, the DC potential does predict the fourfold isomer to be more stable than the three-coordinated one, although the predicted energy difference (1.30 kcal/mol) is larger than the near degeneracy predicted by the high-level electronic structure calculations (0.02 kcal/mol). It should be noted that the energy difference at $T = 0$ K predicted by the DC force field should be compared to the ΔE_0 value predicted by the electronic structure calculations (0.02 kcal/mol) due to the fact that the DC potential can be considered as an effective one which implicitly contains corrections for ZPE, since it is designed to reproduce macroscopic thermodynamic properties using classical MD simulations.

As regards the $n = 5$ cluster, the four-coordinated 4 + 1 cluster is lower in energy than the five- (C_5) and three-coordinated ones (3 + 2) by ~ 3 kcal/mol. Harmonic zero-point energy corrections were found to decrease this energy difference by ~ 0.65 kcal/mol, whereas they practically do not have an effect on the enthalpy at $T = 298$ K. The ordering of the three isomers is well reproduced

by the DFT functionals and the DC force field, although the predicted energy difference between the global 4 + 1 and the local 3 + 2 minima is about one third (DFT) and half (DC) from the electronic structure results. It is interesting to note that the addition of a single water molecule further stabilizes the fourfold coordinated hydroxide structure with respect to the other arrangements (five- and three-coordinated ones) while it destabilizes the three-coordinated (3 + 2) isomer. The above results justify the use of the DC force field for the study of the local structure of aqueous hydroxide.

4. Conclusions

The reorientation of hydroxide ions in water was studied using classical MD simulations with a polarizable force field. Our results for the reorientation relaxation time of OH^- are in a qualitative agreement with the CTTS experiments. The observation that the decay of the orientation correlation functions is well described by a single exponential indicates a successive small angular jump mechanism for the reorientation of both OH^- and H_2O . The calculated results of RDFs (not shown) show excellent agreement with the CPMD and the X-ray diffraction data. Our simulations suggest

two principal solvation structures of OH^- due to the different number of acceptor OH^- HBs. One of the structures has four acceptor HBs from water while the other has five. The population of the four-coordinated HB structure is $\sim 38\%$, while the one for the five-coordinated is $\sim 57\%$ at all temperatures. A weak HB in which the hydroxide acts as the donor is observed in our simulations. No drastic jump of the structure switching was observed.

The angular motion of a hydroxide ion is coupled with the motion of its first solvation shell water molecules. Our results suggest a two-step mechanism of the reorientation of hydroxide ion in water. First, the reorientation of OH^- is initiated by the translation of the water molecules in its first solvation shell. Second, the H atom of the OH^- relaxes to an energetically stable arrangement. The rate-limiting step in this mechanism is the first step, i.e., the translational diffusion of the water molecules in the first solvation shell. The activation energies of reorientation of OH^- and translation of H_2O determined from our simulations are 22.5 and 18.5 kJ/mol, respectively. A subtle difference between the four- and three-coordinated hydroxide was found in small clusters. This competition can be overridden by quenching with the addition of additional solvent molecules, which yields a preference for a four-coordinated hydroxide arrangement in clusters. Nevertheless, our electronic structure results justify the use of the DC force field in the current study and provide a benchmark for the development of accurate force fields that can be used to study the dynamical properties of hydroxide in an aqueous environment in future studies.

Simulating the molecular properties of the surface of aqueous solutions is essential to understanding a wide range of important physical and environmental processes. The factor that gives rise to the thermodynamics properties (i.e., free energy) of ions at interfaces is an area of research at the frontiers of basic chemical physics. From the point of view of molecular simulation, there are many possible ways to model hydroxide anions at an air–water interface. The first and foremost is classical empirical interaction potentials are only able to describe non-covalent bond interactions. A relevant quantity to discern the propensity of ions at interfaces is the free-energy profile of an ion as a function of interfacial depth. The results of our study on the transport property of an OH^- across the water/air interface using the constraint mean force approach shows a flat minimum near the GDS, which indicates that OH^- can be found at the interface, but its free energy is about 1 kcal/mol above the bulk, liquid. We have also carried out simulations of 1 M CsOH salt solution and closely examining the results, we can preliminary conclude that classical simulations show very few hydroxide ions at the aqueous interface, with the majority residing in or inside the bulk liquid. Secondly, the polarizable multi-state empirical valence bond (MS-EVB) approach contains dissociation and allows multiple waters to share protons with the hydroxide anion, but in a computationally efficient way. Lastly, the *ab initio* approaches based on Kohn–Sham density functional theory (DFT) contain the proper physics to describe charge transfer and chemistry but even large scale calculations are limited to short trajectories that will shed little insight on ion-partitioning at the interface. Our work on hydroxide ions at the aqueous interfaces using the above approaches are in progress.

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